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FIRST SEMESTER B.Tech EXAMINATION, DECEMBER 2022

PRINCIPLES OF ELECTRICAL ENGINEERING

Time: 3 Hours **Maximum Marks: 100**

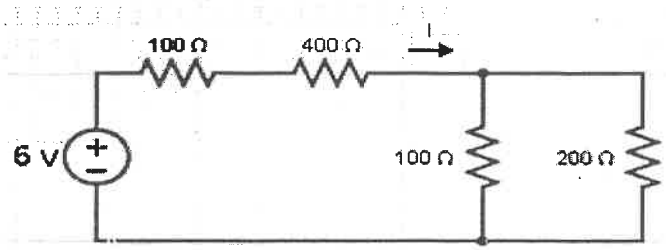
Instructions: **i.** Missing data may be suitably assumed.

PART-A (5X2=10)

ANSWER ALL QUESTIONS

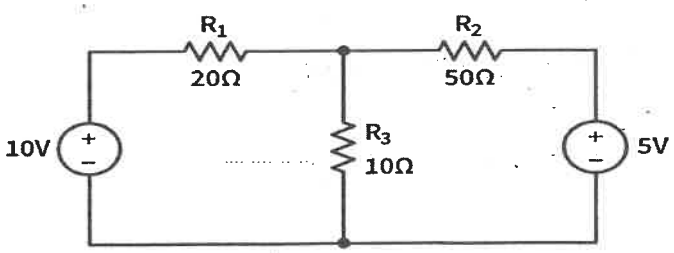
1.	**a.**	Define Kirchhoff's current law and Kirchhoff's voltage law.	L-1	CO1	**02**
	b.	State and explain Thevenin's theorem.	L-1	CO2	**02**
	c.	Define Form factor and Frequency of an alternating quantity.	L-2	CO3	**02**
	d.	Define mutual Induction.	L-1	CO4	**02**
	e.	Explain the necessity of Earthing.	L-2	CO5	**02**

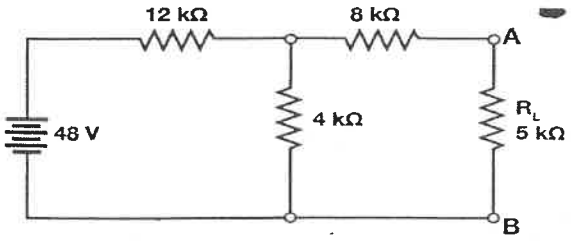
PART-B (6X5=30)

2.	**a.**	Define parallel resistor circuit with circuit diagram and also explain current division in parallel resistor circuit	L- 2	CO1	**05**
		OR			
	b.	The current of 8A is shared between two resistors in the parallel network containing two resistors R1 = 2 Ω and R2 = 4 Ω , calculate the currents in R1 and R2.	L- 2	CO1	**05**
3.	**a.**	State and Explain **i.** Maximum power transfer theorem **ii.** Norton's Theorem	L- 4	CO2	**05**
		OR			
	b.	Find the current I in the circuit shown.  	L- 2	CO2	**05**
4.	**a.**	Derive the expression for voltage, current and power for a single phase AC circuit containing pure resistance	L- 2	CO3	**05**
		OR			
	b.	A coil having a resistance of 7 Ω and an inductance of 31.8 mH is connected to 230V, 50Hz supply. Calculate the circuit current, Power factor and power consumed.	L- 2	CO3	**05**

5.	a.	Define the following parameters i. Self induction ii. Magnetic flux iii. Magnetic flux density iv. Transformation ratio v. Electric field strength	L- 2	CO4	05
		OR			
	b.	Explain how the total capacitance of the circuit varies when the capacitors are connected in series?	L- 2	CO4	05
6.	a.	Mention the types of electrical wiring and explain any one type in detail.	L- 4	CO5	05
		OR			
	b.	Find the total capacitance for three capacitors connected in series, given their individual capacitances are 1 μF, 5 μF, and 8 μF.	L- 4	CO5	05
7.	a.	Explain the working of a Piezo electric transducer	L- 2	CO6	05
		OR			
	b.	What is thermocouple? Explain its working principle.	L- 2	CO6	05

PART-C (6X10=60)

8.	a.	Derive the expression for voltage, current and power for a single phase AC circuit containing R-L connected in series. Draw the necessary waveform and phasor diagrams.	L- 2	CO1	10
		OR			
	b.	Determine the current flowing through R1, R2 and R3 in the given circuit  	L- 2	CO1	10
9.	a.	A 230 V, 50 Hz ac supply is applied to a coil of 0.06 H inductance and 2.5 Ohm resistance connected in series with a 6.8 μF capacitor. Calculate; i. Impedance ii. Current iii. Phase angle between current and voltage iv. Power factor v. Power consumed	L- 4	CO2	10
		OR			

	b.	Find V_{TH}, R_{TH} and the load current I_L flowing through load and load voltage across the load resistor in the circuit below using Thevenin's Theorem. 	L- 4	CO2	**10**
10.	**a.**	Derive the relationship between phase voltage and line voltage in a star connected 3 phase balanced AC circuit.	L- 2	CO3	**10**
		OR			
	b.	Derive the expression for voltage, current and power for a single phase AC circuit containing pure inductance. Draw the necessary waveforms and phasor diagrams.	L- 2	CO3	**10**
11.	**a.**	Derive the expression for energy stored in a charged capacitor.	L- 2	CO4	**10**
		OR			
	b.	Derive the expression for EMF induced in primary and secondary winding of a transformer with neat circuit diagram.	L- 4	CO4	**10**
12.	**a.**	List the types of Earthing. Explain any one type of Earthing in detail.	L- 2	CO5	**10**
		OR			
	b.	Explain the working of MCB and Fuse.	L- 2	CO5	**10**
13.	**a.**	With a neat circuit diagram, explain the principle of operation of a induction type wattmeter.	L- 4	CO6	**10**
		OR			
	b.	With a neat circuit diagram, explain the principle of operation of an ammeter.	L- 4	CO6	**10**